



guests. But then, following an argument with Clement about the cost of tools, work on the prototype stopped.

A decade later, Babbage refined his designs to create the Difference Engine No.2. This would need only a third of the parts of the earlier prototype and would be able to calculate numbers up to 31 digits long. However, this time no funding was provided, and the machine was not

constructed in Babbage's lifetime. More than a century later, in 1991, the London Science Museum completed his vision by building a full-scale working model.

The Analytical Engine

In 1833, Babbage started a new project, his Analytical Engine. A more advanced design than the Difference Engine, it would be able to add, subtract, multiply,

**THE BRITISH
GOVERNMENT
CONTRIBUTED
\$22,100
TO THE
CONSTRUCTION
OF THE
DIFFERENCE
ENGINE NO.1**



JOHN HERSCHEL

The mathematician and astronomer John Herschel produced the first global survey of the night sky, named the moons of Saturn and Uranus, and coined the word "photography."



Herschel (1792–1871) studied mathematics at Cambridge University with Babbage and co-founded the Analytical Society with him. He also helped found the Royal Astronomical Society and served as its president three times. He discovered more than 500 nebulae and 3,000 double stars in the northern hemisphere. In 1821, Herschel was awarded the Royal Society's Copley Medal for his contributions to mathematical analysis. In 1834, he went to South Africa, set up the first observatory in the southern hemisphere, and completed his catalog of the Earth's skies. He was also the first to capture a permanent photographic image, in 1839.

**THE ANALYTICAL
ENGINE
COULD STORE
1,000
50-DIGIT
NUMBERS:
MORE THAN ANY
PRE-1960S'
COMPUTER**